Matthew Curley

03/03/26

1.

```
MariaDB [mynewpaltz]> SELECT s.firstName, s.lastName, s.ssn FROM StudentTable s JOIN EnrollmentTable e ON s.ssn = e.ssn GROUP BY s.ssn, s.firstName,
 s.lastName HAVING COUNT(e.courseID) = 5;
+-----------+----------+-----------+
| firstName | lastName | ssn       |
+-----------+----------+-----------+
| Kevin     | Green    | 100000041 |
| Michelle  | Adams    | 100000042 |
| Brian     | Nelson   | 100000043 |
| Carol     | Baker    | 100000044 |
| Edward    | Hall     | 100000045 |
| Ronald    | Campbell | 100000047 |
| Melissa   | Mitchell | 100000048 |
| Timothy   | Carter   | 100000049 |
| Deborah   | Roberts  | 100000050 |
| Stephanie | Phillips | 100000052 |
| Jeffrey   | Evans    | 100000053 |
| Rebecca   | Turner   | 100000054 |
| Ryan      | Diaz     | 100000055 |
| Laura     | Parker   | 100000056 |
| Sharon    | Edwards  | 100000058 |
| Gary      | Collins  | 100000059 |
| Cynthia   | Reyes    | 100000060 |
+-----------+----------+-----------+
17 rows in set (0.013 sec)
```

2.

```
MariaDB [mynewpaltz]> SELECT s.firstName, s.lastName, s.ssn FROM StudentTable s JOIN EnrollmentTable e ON s.ssn = e.ssn GROUP BY s.ssn, s.firstName,
 s.lastName HAVING COUNT(e.courseID) = 4;
+-----------+----------+-----------+
| firstName | lastName | ssn       |
+-----------+----------+-----------+
| Amanda    | Rivera   | 100000046 |
| Jacob     | Cruz     | 100000057 |
+-----------+----------+-----------+
2 rows in set (0.001 sec)
```

3.

```
MariaDB [mynewpaltz]> SELECT s.firstName, s.lastName, s.ssn FROM StudentTable s JOIN EnrollmentTable e ON s.ssn = e.ssn GROUP BY s.ssn, s.firstName,
 s.lastName HAVING COUNT(e.courseID) = 3;
+-------------+-----------+-----------+
| firstName   | lastName  | ssn       |
+-------------+-----------+-----------+
| James       | Smith     | 100000001 |
| Mary        | Johnson   | 100000002 |
| Robert      | Williams  | 100000003 |
| Patricia    | Brown     | 100000004 |
| John        | Jones     | 100000005 |
| Jennifer    | Garcia    | 100000006 |
| Michael     | Miller    | 100000007 |
| Linda       | Davis     | 100000008 |
| David       | Rodriguez | 100000009 |
| Elizabeth   | Martinez  | 100000010 |
| William     | Hernandez | 100000011 |
| Barbara     | Lopez     | 100000012 |
| Richard     | Gonzalez  | 100000013 |
| Susan       | Wilson    | 100000014 |
| Joseph      | Anderson  | 100000015 |
| Jessica     | Thomas    | 100000016 |
| Thomas      | Taylor    | 100000017 |
| Sarah       | Moore     | 100000018 |
| Christopher | Jackson   | 100000019 |
| Karen       | Martin    | 100000020 |
+-------------+-----------+-----------+
20 rows in set (0.001 sec)
```

4.

```
MariaDB [mynewpaltz]> SELECT s.firstName, s.lastName, s.ssn FROM StudentTable s JOIN EnrollmentTable e ON s.ssn = e.ssn GROUP BY s.ssn, s.firstName,
 s.lastName HAVING COUNT(e.courseID) = 2;
+-----------+-----------+-----------+
| firstName | lastName  | ssn       |
+-----------+-----------+-----------+
| Nicholas  | Stewart   | 100000061 |
| Kathleen  | Morris    | 100000062 |
| Eric      | Morales   | 100000063 |
| Amy       | Murphy    | 100000064 |
| Stephen   | Cook      | 100000065 |
| Shirley   | Rogers    | 100000066 |
| Jonathan  | Gutierrez | 100000067 |
| Angela    | Ortiz     | 100000068 |
| Larry     | Morgan    | 100000069 |
| Helen     | Cooper    | 100000070 |
| Justin    | Peterson  | 100000071 |
| Anna      | Bailey    | 100000072 |
| Scott     | Reed      | 100000073 |
| Brenda    | Kelly     | 100000074 |
| Brandon   | Howard    | 100000075 |
| Pamela    | Ramos     | 100000076 |
| Frank     | Kim       | 100000077 |
| Nicole    | Foster    | 100000078 |
| Benjamin  | Alvarado  | 100000079 |
| Emma      | Washington | 100000080 |
| Gregory   | Butler    | 100000081 |
| Samantha  | Simmons   | 100000082 |
| Samuel    | Foster    | 100000083 |
| Katherine | Gonzales  | 100000084 |
| Raymond   | Bryant    | 100000085 |
| Christine | Alexander | 100000086 |
| Patrick   | Russell   | 100000087 |
| Debra     | Griffin   | 100000088 |
| Alexander | Diaz      | 100000089 |
| Rachel    | Hayes     | 100000090 |
```

5.

```
MariaDB [mynewpaltz]> SELECT s.firstName, s.lastName, s.ssn FROM StudentTable s JOIN EnrollmentTable e ON s.ssn = e.ssn GROUP BY s.ssn, s.firstName,
 s.lastName HAVING COUNT(e.courseID) = 1;
+-----------+----------+-----------+
| firstName | lastName | ssn       |
+-----------+----------+-----------+
| Charles   | Lee      | 100000021 |
| Nancy     | Perez    | 100000022 |
| Matthew   | Thompson | 100000023 |
| Lisa      | White    | 100000024 |
| Anthony   | Harris   | 100000025 |
| Betty     | Sanchez  | 100000026 |
| Mark      | Clark    | 100000027 |
| Margaret  | Ramirez  | 100000028 |
| Donald    | Lewis    | 100000029 |
| Sandra    | Robinson | 100000030 |
| Steven    | Walker   | 100000031 |
| Ashley    | Young    | 100000032 |
| Paul      | Allen    | 100000033 |
| Dorothy   | King     | 100000034 |
| Andrew    | Wright   | 100000035 |
| Kimberly  | Scott    | 100000036 |
| Joshua    | Torres   | 100000037 |
| Emily     | Nguyen   | 100000038 |
| Kenneth   | Hill     | 100000039 |
| Donna     | Flores   | 100000040 |
+-----------+----------+-----------+
20 rows in set (0.001 sec)
```

6.

```
MariaDB [mynewpaltz]> SELECT s.firstName, s.lastName, s.ssn FROM StudentTable s LEFT JOIN EnrollmentTable e ON s.ssn = e.ssn WHERE e.courseID IS NUL
L;
+-----------+----------+-----------+
| firstName | lastName | ssn       |
+-----------+----------+-----------+
| Jason     | Gomez    | 100000051 |
| Jack      | Myers    | 100000091 |
+-----------+----------+-----------+
2 rows in set (0.004 sec)
```

7.

```
MariaDB [mynewpaltz]> SELECT subjectID, COUNT(*) AS total_4_credit_courses FROM CourseTable WHERE numOfCredits = 4 GROUP BY subjectID;
+-----------+------------------------+
| subjectID | total_4_credit_courses |
+-----------+------------------------+
| CS        |                      2 |
| ENGR      |                      2 |
| MATH      |                      2 |
+-----------+------------------------+
3 rows in set (0.004 sec)
```

8.

```
MariaDB [mynewpaltz]> SELECT subjectID, COUNT(*) AS total_3_credit_courses FROM CourseTable WHERE numOfCredits = 3 GROUP BY subjectID;
+-----------+------------------------+
| subjectID | total_3_credit_courses |
+-----------+------------------------+
| CS        |                      3 |
| ENGL      |                      5 |
| ENGR      |                      2 |
| MATH      |                      3 |
+-----------+------------------------+
4 rows in set (0.001 sec)
```

9.

```
MariaDB [mynewpaltz]> SELECT subjectID, COUNT(*) AS total_2_credit_courses FROM CourseTable WHERE numOfCredits = 2 GROUP BY subjectID;
+-----------+-----------------------+
| subjectID | total_2_credit_courses |
+-----------+-----------------------+
| ENGR      |                     1 |
+-----------+-----------------------+
1 row in set (0.001 sec)
```

10. One 2 credit course

```
MariaDB [mynewpaltz]> SELECT subjectID, COUNT(*) AS total_2_credit_courses FROM CourseTable WHERE numOfCredits = 2 GROUP BY subjectID;
+-----------+-----------------------+
| subjectID | total_2_credit_courses |
+-----------+-----------------------+
| ENGR      |                     1 |
+-----------+-----------------------+
1 row in set (0.001 sec)
```

Zero 1 credit courses

```
MariaDB [mynewpaltz]> SELECT COUNT(*) AS total_1_credit_courses FROM CourseTable WHERE numOfCredits = 1;
+-------------------------+
| total_1_credit_courses |
+-------------------------+
|                       0 |
+-------------------------+
1 row in set (0.002 sec)
```

11.

```
MariaDB [mynewpaltz]> SELECT s.firstName, s.lastName, e.courseID, e.grade FROM StudentTable s JOIN EnrollmentTable e ON s.ssn = e.ssn ORDER BY s.las
tName;
+-------------+------------+----------+-------+
| firstName   | lastName   | courseID | grade |
+-------------+------------+----------+-------+
| Michelle    | Adams      |        6 | B     |
| Michelle    | Adams      |        7 | A     |
| Michelle    | Adams      |        8 | C     |
| Michelle    | Adams      |        9 | B     |
| Michelle    | Adams      |       10 | A     |
| Christine   | Alexander  |       10 | C     |
| Christine   | Alexander  |       15 | A     |
| Paul        | Allen      |       13 | A     |
| Benjamin    | Alvarado   |        3 | A     |
| Benjamin    | Alvarado   |        8 | C     |
| Joseph      | Anderson   |        5 | A     |
| Joseph      | Anderson   |       15 | A     |
| Joseph      | Anderson   |       20 | B     |
| Anna        | Bailey     |        1 | B     |
| Anna        | Bailey     |       16 | B     |
| Carol       | Baker      |       16 | A     |
| Carol       | Baker      |       17 | C     |
| Carol       | Baker      |       18 | B     |
| Carol       | Baker      |       19 | A     |
| Carol       | Baker      |       20 | B     |
| Patricia    | Brown      |        4 | B     |
| Patricia    | Brown      |        9 | B     |
| Patricia    | Brown      |       14 | A     |
| Raymond     | Bryant     |        9 | A     |
| Raymond     | Bryant     |       14 | C     |
| Gregory     | Butler     |        5 | B     |
| Gregory     | Butler     |       10 | B     |
| Ronald      | Campbell   |       11 | C     |
| Ronald      | Campbell   |       13 | B     |
| Ronald      | Campbell   |       15 | A     |
```

12.

```
MariaDB [mynewpaltz]> SELECT c.subjectID, c.courseNumber, COUNT(e.ssn) AS total_students FROM CourseTable c LEFT JOIN EnrollmentTable e ON c.courseI
D = e.courseID WHERE c.courseID = 10 GROUP BY c.subjectID, c.courseNumber;
+-----------+--------------+----------------+
| subjectID | courseNumber | total_students |
+-----------+--------------+----------------+
| CS        |          450 |             13 |
+-----------+--------------+----------------+
1 row in set (0.005 sec)
```

13.

```
MariaDB [mynewpaltz]> SELECT c.courseID, COUNT(e.ssn) AS total_students FROM CourseTable c LEFT JOIN EnrollmentTable e ON c.courseID = e.courseID GR
OUP BY c.courseID ORDER BY c.courseID;
+----------+----------------+
| courseID | total_students |
+----------+----------------+
|        1 |             13 |
|        2 |             13 |
|        3 |             12 |
|        4 |             14 |
|        5 |             13 |
|        6 |             11 |
|        7 |             13 |
|        8 |             13 |
|        9 |             12 |
|       10 |             13 |
|       11 |             14 |
|       12 |             13 |
|       13 |             14 |
|       14 |             14 |
|       15 |             12 |
|       16 |             12 |
|       17 |             12 |
|       18 |             12 |
|       19 |             11 |
|       20 |             10 |
+----------+----------------+
20 rows in set (0.003 sec)
```

14.

```
MariaDB [mynewpaltz]> SELECT s.firstName, s.lastName, SUM(c.numOfCredits) AS total_credits FROM StudentTable s JOIN EnrollmentTable e ON s.ssn = e.s
sn JOIN CourseTable c ON e.courseID = c.courseID GROUP BY s.ssn, s.firstName, s.lastName ORDER BY s.lastName;
+-------------+------------+---------------+
| firstName   | lastName   | total_credits |
+-------------+------------+---------------+
| Michelle    | Adams      |            17 |
| Christine   | Alexander  |             6 |
| Paul        | Allen      |             3 |
| Benjamin    | Alvarado   |             7 |
| Joseph      | Anderson   |            10 |
| Anna        | Bailey     |             6 |
| Carol       | Baker      |            16 |
| Patricia    | Brown      |             9 |
| Raymond     | Bryant     |             6 |
| Gregory     | Butler     |             6 |
| Ronald      | Campbell   |            16 |
| Timothy     | Carter     |            16 |
| Mark        | Clark      |             3 |
| Heather     | Cole       |             7 |
| Gary        | Collins    |            16 |
| Stephen     | Cook       |             6 |
| Helen       | Cooper     |             7 |
| Jacob       | Cruz       |            12 |
| Linda       | Davis      |            10 |
| Ryan        | Diaz       |            15 |
| Alexander   | Diaz       |             6 |
| Sharon      | Edwards    |            16 |
| Jeffrey     | Evans      |            16 |
| Donna       | Flores     |             4 |
| Janet       | Ford       |             6 |
| Samuel      | Foster     |             6 |
| Nicole      | Foster     |             7 |
| Jennifer    | Garcia     |            10 |
| Katherine   | Gonzales   |             7 |
| Richard     | Gonzalez   |             9 |
```